Lecture 6
Intermolecular Forces and Liquids

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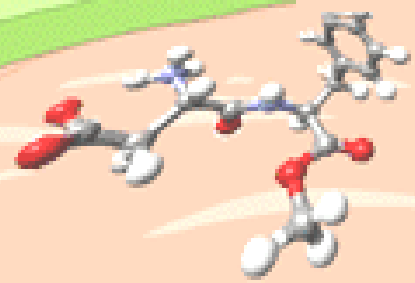
قسم الكيمياء- كلية العلوم- جامعة المنيا



Intermolecular Forces

- Intramolecular forces: The forces of attraction that exist between bonds **within** a molecule (e.g. ionic bond and covalent bonds).
- Intermolecular forces: The forces of attraction that exist between **between** molecules.

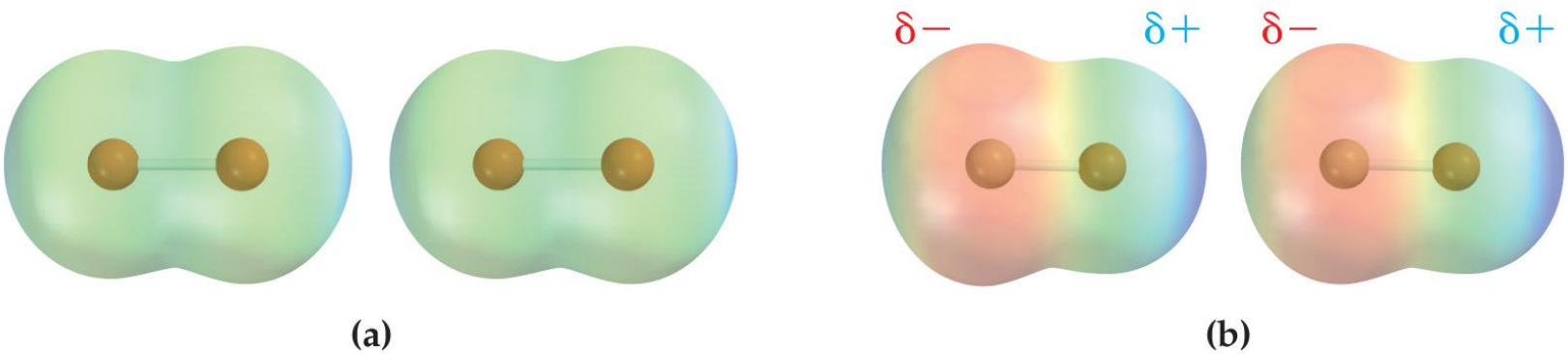
Types of Intermolecular Forces:

1. **Dispersion Forces**
2. **Dipole-Dipole Forces**
3. **Hydrogen Bonding**
4. **Ion-Dipole Forces**

- The strength of the intermolecular forces between the molecules or atoms that compose a substance determines the state—solid, liquid, or gas—of the substance at room temperature.
- Strong intermolecular forces tend to result in liquids and solids (with high melting and boiling points).
- Weak intermolecular forces tend to result in gases (with low melting and boiling points).

Dispersion Forces (also called the *London force*)

- Dispersion forces are caused by fluctuations in the electron distribution within molecules or atoms.
- the electron distribution in a nonpolar molecule is symmetrical. (b) At any instant, it may be unsymmetrical, resulting in a temporary polarity that can attract neighboring molecules.
- Since all atoms and molecules have electrons, they all have dispersion forces.
- The tendency of an electron cloud to distort in this way is called polarizability.



Dispersion Forces

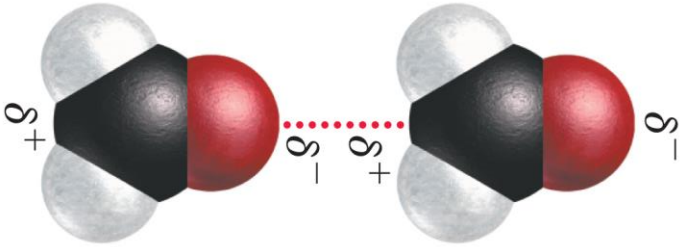
- A larger electron cloud results in a greater dispersion force because the electrons are held less tightly by the nucleus and therefore can polarize more easily.
- To *polarize* means to form a dipole moment.
- While molar mass alone does not determine the magnitude of the dispersion force, it can be used as a guide when comparing dispersion forces within a family of similar elements or compounds

Noble gas boing points

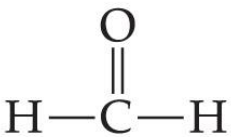
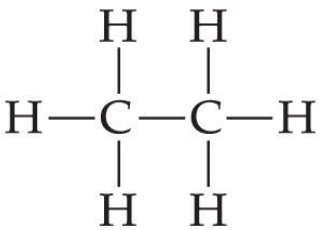
Noble Gas	Molar Mass (g/mol)	Boiling Point (K)
He	4.00	4.2 K
Ne	20.18	27 K
Ar	39.95	87 K
Kr	83.80	120 K
Xe	131.29	165 K

Dipole–Dipole Force

- The **dipole–dipole force** exists in all polar molecules.
- Polar molecules have **permanent dipoles** that interact with the permanent dipoles of neighboring molecules.
- The positive end of one permanent dipole is attracted to the negative end of another; this attraction is the dipole–dipole force.
- Remember that all molecules (including polar ones) have dispersion forces.
- In addition, polar molecules have dipole–dipole forces.
- These additional attractive forces raise their melting and boiling points relative to nonpolar molecules of similar molar mass.
- Molecules such as formaldehyde are polar and therefore have a permanent dipole.



Effect of Difference in Polarity on Melting and Boiling Points for Two Compounds of Similar Molar Mass

Name	Formula	Molar mass (g/mol)	Structure	bp (°C)	mp (°C)
Formaldehyde	CH_2O	30.0		−19.5	−92
Ethane	C_2H_6	30.1		−88	−172

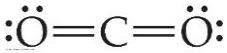
Miscibility—a Liquid's Ability to Mix with Another Liquid without Separating into Two Phases

- In general, polar liquids are miscible with other polar liquids but are not miscible with nonpolar liquids.
- For example, water, a polar liquid, is not miscible with pentane (C_5H_{12}), a nonpolar liquid. Similarly, water and oil (also nonpolar) do not mix.

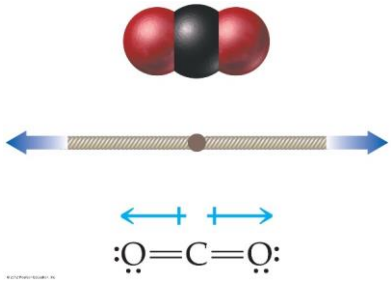
Polar Bonds and Polar Molecules

- Does the presence of one or more polar bonds in a molecule always result in a polar molecule? The answer is NO.
- A **polar molecule** is one with polar bonds that add together—they do not cancel each other—to form a net dipole moment.
- When a diatomic molecule contains a polar bond, then the molecule is polar.
- For molecules with more than two atoms, it is more difficult to tell polar molecules from nonpolar ones because two or more polar bonds may cancel one another.

Consider carbon dioxide:



- Each *bond* is polar because the difference in electronegativity between oxygen and carbon is 1.0.
- CO_2 has a linear geometry, the dipole moment of one bond completely cancels the dipole moment of the other, and the *molecule* is nonpolar.

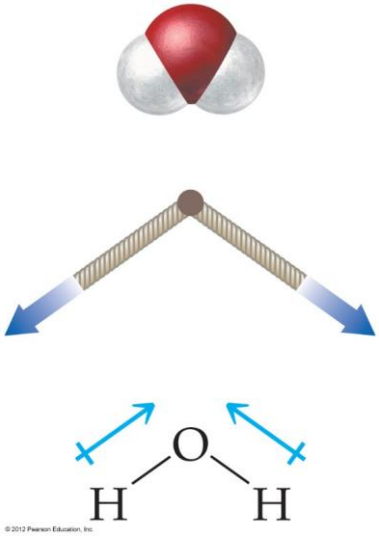


Vector Notation for Dipole Moments

- We can represent polar bonds with arrows (vectors) that point in the direction of the negative pole and have a plus sign at the positive pole (as just shown for carbon dioxide).
- If the arrows (vectors) point in exactly opposing directions as in carbon dioxide, the dipole moments cancel.
- In the vector representation of a dipole moment, the vector points *in the direction of the atom with the partial negative charge*.

Consider water (H_2O):

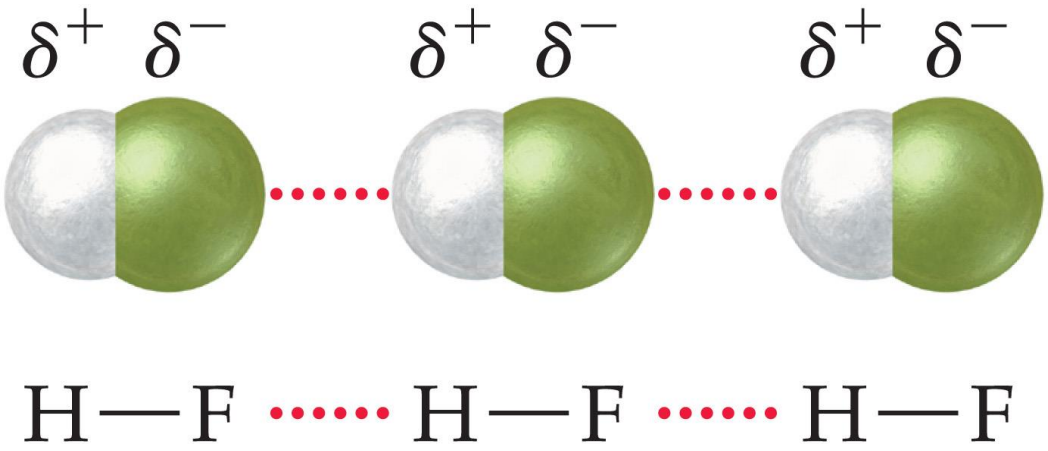
- Each *bond* is polar because the difference in electronegativity between oxygen and hydrogen is 1.4.
- Water has two dipole moments that do not cancel, and the *molecule* is polar.



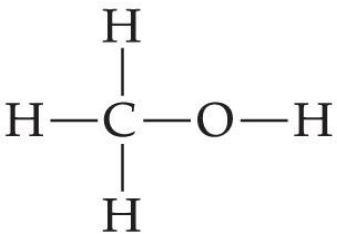
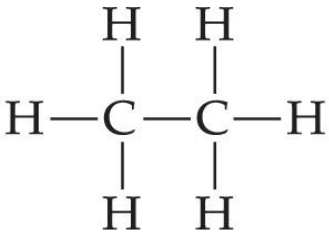
Hydrogen Bonding

The Intermolecular Attraction of a Hydrogen Atom to an Electronegative Atom is Called a *Hydrogen Bond*

- Polar molecules containing hydrogen atoms bonded directly to fluorine, oxygen, or nitrogen exhibit an additional intermolecular force called a **hydrogen bond.**
- HF, NH_3, and H_2O all undergo hydrogen bonding.
- A hydrogen bond is a sort of ***super*** dipole–dipole force.
- In HF, the hydrogen on each molecule is strongly attracted to the fluorine *on its neighbor.*

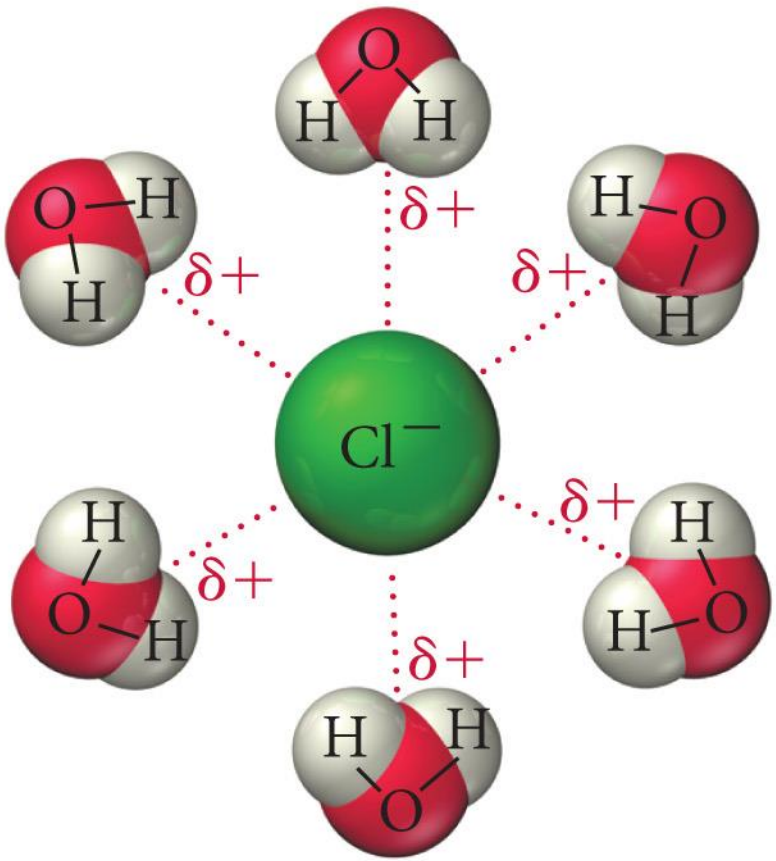


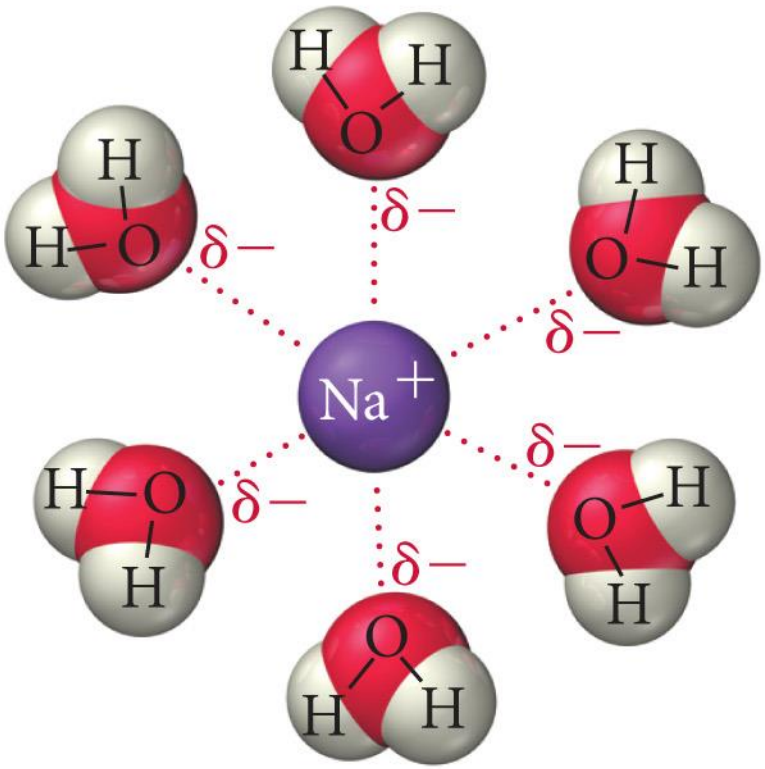
Effect of Hydrogen Bonding on Melting and Boiling Points for Two Compounds of Similar Molar Mass

Name	Formula	Molar mass (g/mol)	Structure	bp (°C)	mp (°C)
Methanol	CH_3OH	32.0		64.7	−97.8
Ethane	C_2H_6	30.1		−88	−172

Ion–Dipole Forces

Ion–dipole forces exist between Na^+ and the negative ends of H_2O molecules and between Cl^- and the positive ends of H_2O molecules.





Thermochemistry

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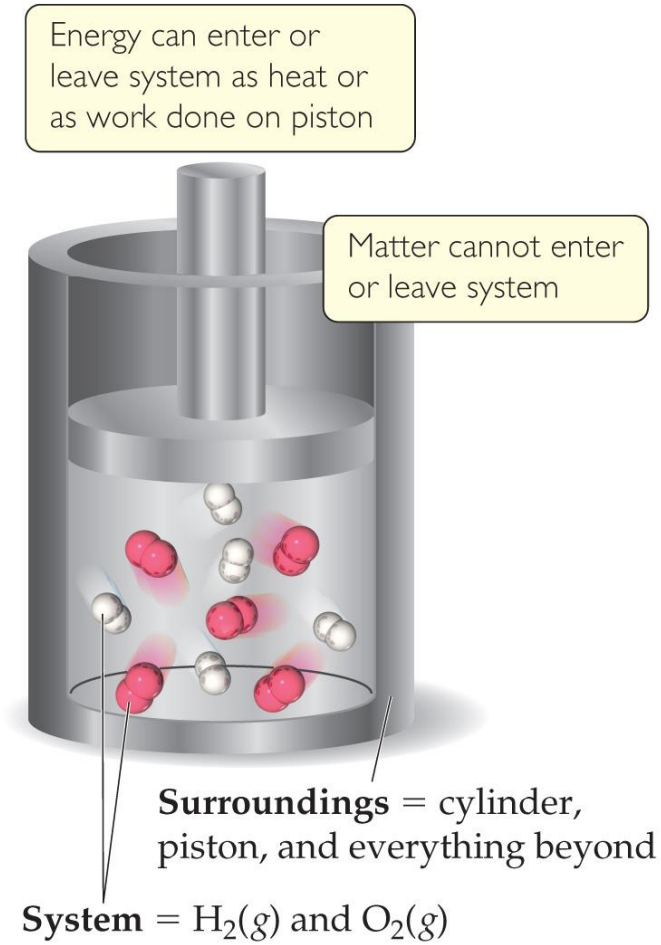
Definitions: System and Surroundings

- The **system** includes the molecules we want to study (here, the hydrogen and oxygen molecules).
- The **surroundings** are everything else (here, the cylinder and piston).



Work

- Energy used to move an object over some distance is **work**:

$$w = F \times d$$

where w is work, F is the force, and d is the distance over which the force is exerted.

Heat

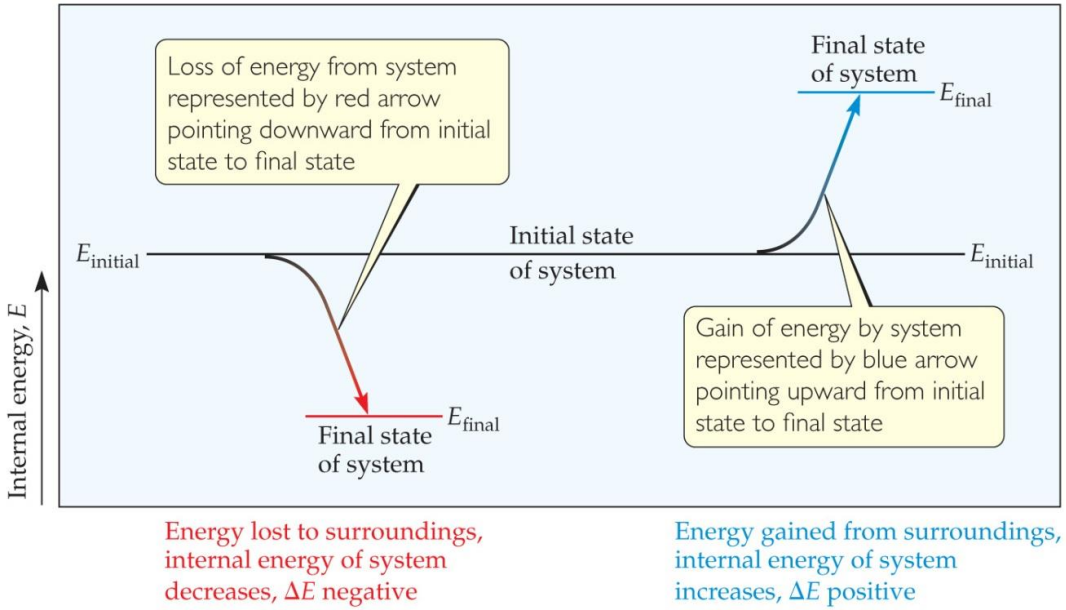
- Energy can also be transferred as heat.
- Heat flows from warmer objects to cooler objects.

First Law of Thermodynamics

- the total energy of the universe is a constant; if the system loses energy, it must be gained by the surroundings, and vice versa.

Internal Energy

- The **internal energy** of a system is the sum of all kinetic and potential energies of all components of the system; we call it *E.*
- We generally don't know E, only how it CHANGES (**ΔE**).



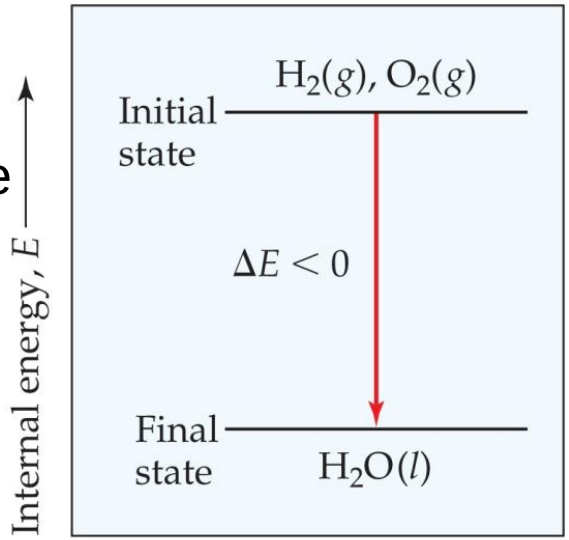
Internal Energy

By definition, the change in internal energy, ΔE, is the final energy of the system minus the initial energy of the system:

$$\Delta E = E_{final} - E_{initial}$$

- **If $\Delta E > 0$, $E_{final} > E_{initial}$**
  - Therefore, the system *absorbed* energy from the surroundings.

- **If $\Delta E < 0$, $E_{final} < E_{initial}$**
  - Therefore, the system *released* energy to the surroundings.

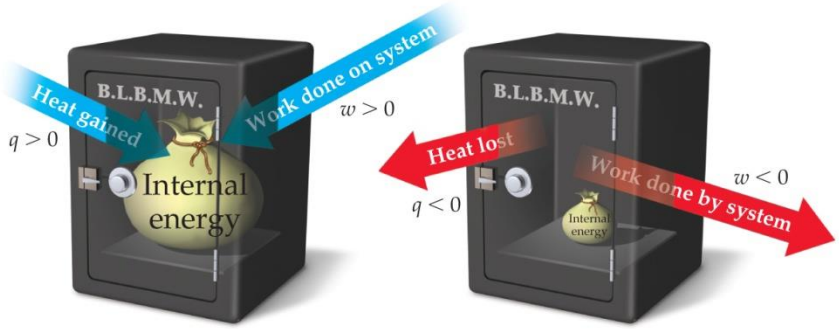


$E_{initial}$ greater than E_{final}; therefore, energy is released from system to surroundings during reaction and $\Delta E < 0$

Changes in Internal Energy

- When energy is exchanged between the system and the surroundings, it is exchanged as either heat (q) or work (w).
- That is, $\Delta E = q + w$.

System is interior of vault



Energy deposited into system
$\Delta E > 0$

Energy withdrawn from system
$\Delta E < 0$

ΔE, q, w, and Their Signs

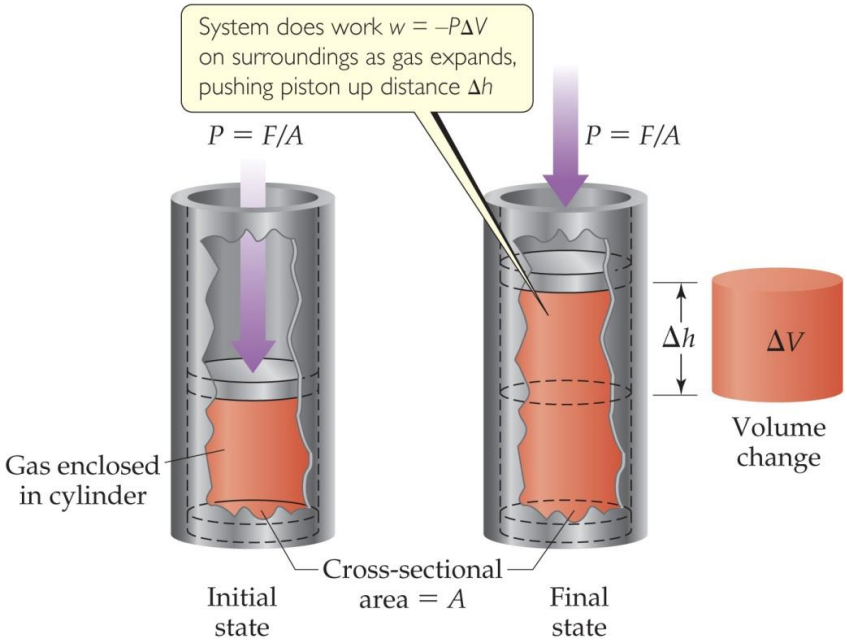
Sign Conventions for q, w, and ΔE		
For q	+ means system *gains* heat	− means system *loses* heat
For w	+ means work done *on* system	− means work done *by* system
For ΔE	+ means *net gain* of energy by system	− means *net loss* of energy by system

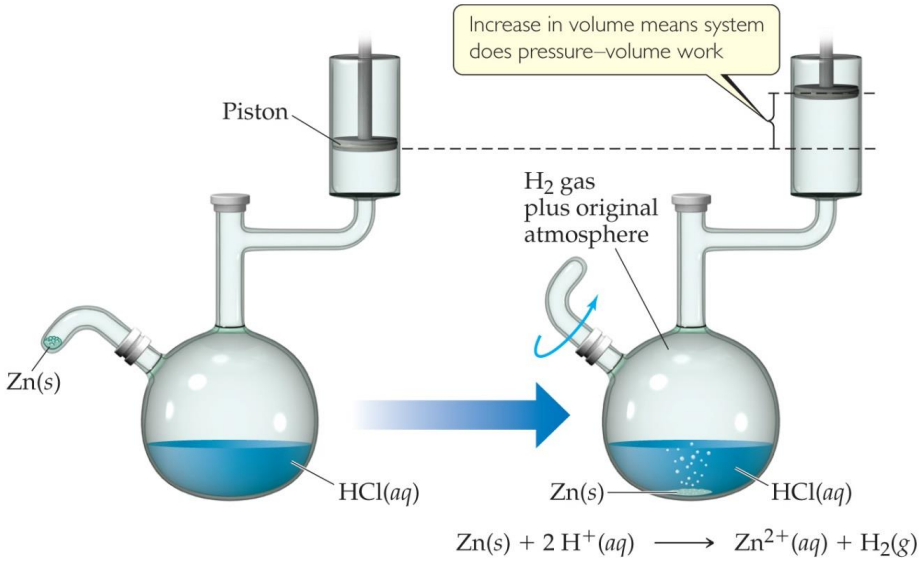
Work

- Usually the only work done by chemical or physical change is the mechanical work associated with a change in volume of gas.

- We can measure the work done by the gas if the reaction is done in a vessel that has been fitted with a piston:

$$w = -P\Delta V$$





Enthalpy

- **Enthalpy** is the internal energy plus the product of pressure and volume:

$$H = E + PV$$

- If a process takes place at constant pressure (and we usually work at atmospheric pressure) and the only work done is this pressure–volume work, we can account for heat flow during the process by measuring the *enthalpy* of the system.

- When the system changes at constant pressure, the change in enthalpy, ΔH, is

$$\Delta H = \Delta(E + PV)$$

- This can be written

$$\Delta H = \Delta E + P\Delta V$$

Enthalpy

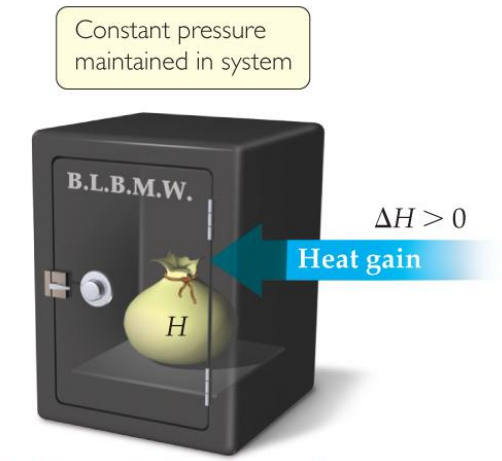
- Since $\Delta E = q + w$ and $w = -P\Delta V$, we can substitute these into the enthalpy expression:

$$\Delta H = \Delta E + P\Delta V$$

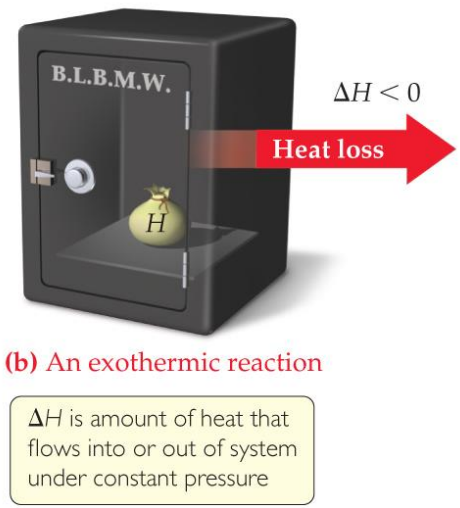
$$\Delta H = (q + w) - w$$

$$\Delta H = q$$

- So, at constant pressure, the change in enthalpy is the heat gained or lost.



(a) An endothermic reaction



(b) An exothermic reaction

A process is endothermic when ΔH is positive.

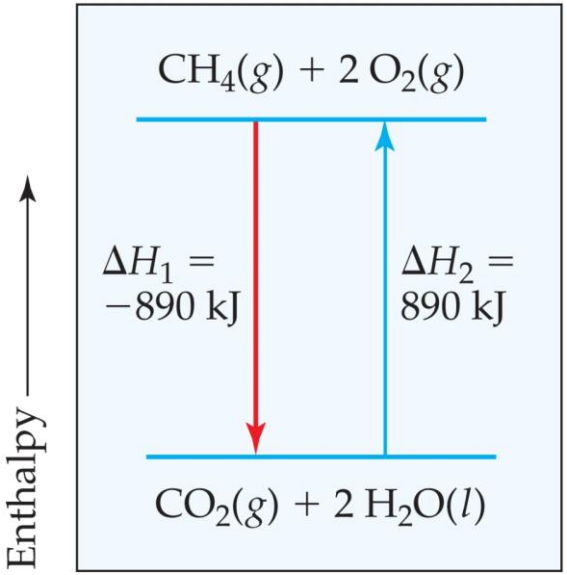
A process is exothermic when ΔH is negative.

Enthalpy of Reaction

- The change in enthalpy, ΔH, is the enthalpy of the products minus the enthalpy of the reactants:

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

- This quantity, ΔH, is called the **enthalpy of reaction**, or the **heat of reaction**.



The Truth about Enthalpy

1. ΔH for a reaction in the forward direction is equal in size, but opposite in sign, to ΔH for the reverse reaction.

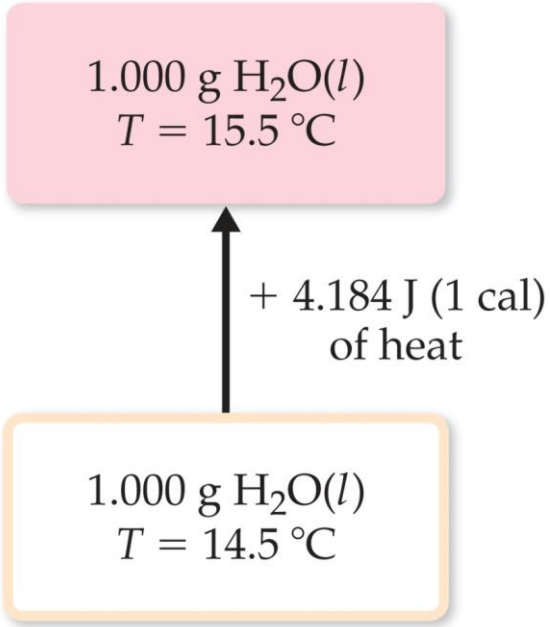
2. ΔH for a reaction depends on the state of the products and the state of the reactants.

Heat Capacity and Specific Heat

- The amount of energy required to raise the temperature of a substance by 1 K (1 °C) is its **heat capacity (C)**, usually given for one mole of the substance (**molar heat capacity**).

$$c = \frac{q}{\Delta T}$$

- We define **specific heat capacity** (C_s) as the amount of energy required to raise the temperature of **1 g** of a substance by 1 K (or 1 °C).



Specific heat, then, is

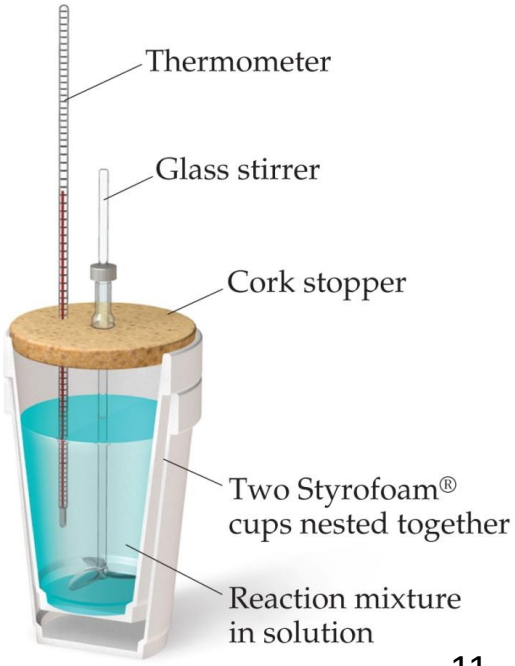
$$\text{Specific heat} = \frac{\text{heat transferred}}{\text{mass} \times \text{temperature change}}$$

$$C_s = \frac{q}{m \times \Delta T}$$

Calorimetry

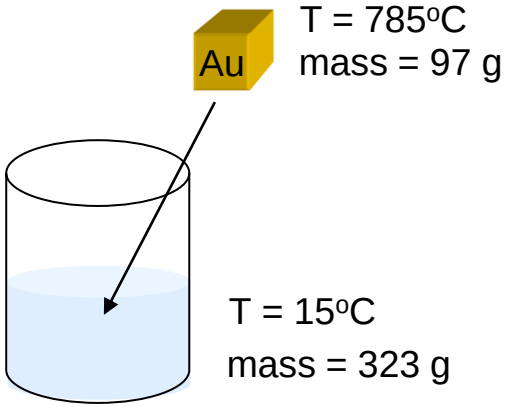
Constant Pressure Calorimetry

- Since we cannot know the exact enthalpy of the reactants and products, we measure ΔH through **calorimetry**, the measurement of heat flow.
- The instrument used to measure heat flow is called a **calorimeter**.
- By carrying out a reaction in aqueous solution in a simple calorimeter, the heat change for the system can be found by measuring the heat change for the water in the calorimeter.
- The specific heat for water is well known (4.184 J/g·K).
- We can calculate ΔH for the reaction with this equation: $q_{soln} = C_s \times m_{soln} \times \Delta T = -q_{rxn}$



Solved problems

1) A 97 g sample of gold at 785°C is dropped into 323 g of water, which has an initial temperature of 15°C. If gold has a specific heat of 0.129 J/g°C, what is the final temperature of the mixture? Assume that the gold experiences no change in state of matter.



$$-\text{ LOSE heat} = \text{GAIN heat}$$

$$-[(C_{p,Au})(\text{mass})(\Delta T)] = (C_{p,H_2O})(\text{mass})(\Delta T)$$

Drop Units:

$$-[(0.129\ \text{J/g}^{\circ}\text{C})(97\ \text{g})(T_f - 785^{\circ}\text{C})] = (4.184\ \text{J/g}^{\circ}\text{C})(323\ \text{g})(T_f - 15^{\circ}\text{C})]$$

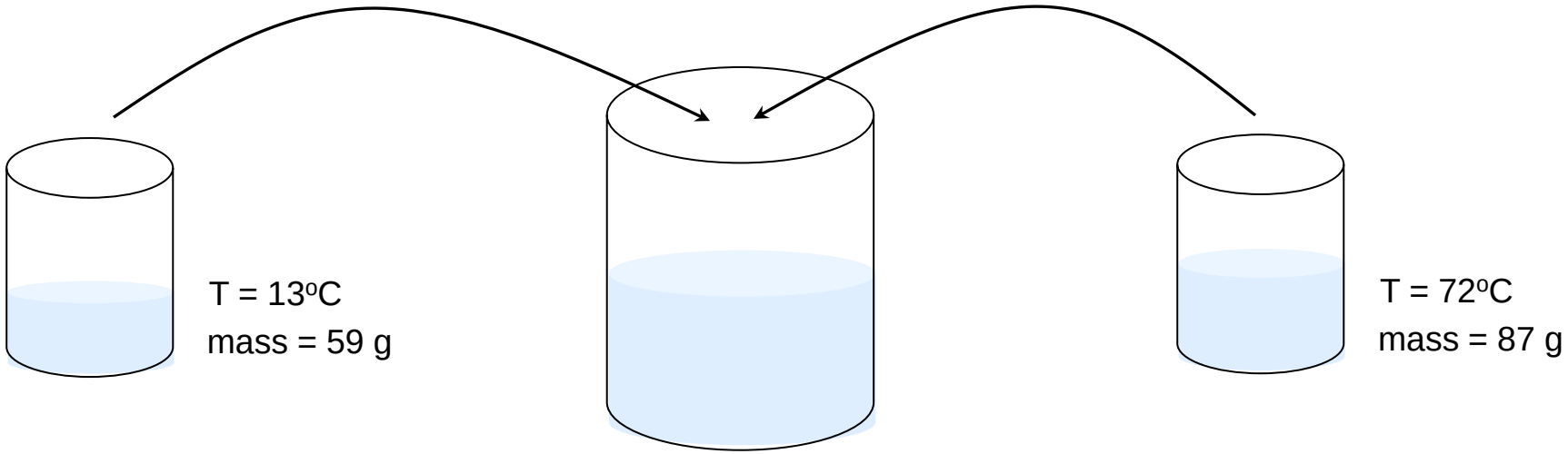
$$-[(12.5)(T_f - 785^{\circ}\text{C})] = (1.35 \times 10^3)(T_f - 15^{\circ}\text{C})]$$

$$-12.5\ T_f + 9.82 \times 10^3 = 1.35 \times 10^3\ T_f - 2.02 \times 10^4$$

$$3 \times 10^4 = 1.36 \times 10^3\ T_f$$

$$T_f = 22.1^{\circ}\text{C}$$

2) If 59 g of water at 13°C are mixed with 87 g of water at 72°C, find the final temperature of the system.



$$-\text{ LOSE heat} = \text{GAIN heat}$$

$$-[(C_{p,H_2O})(\text{mass})(\Delta T)] = (C_{p,H_2O})(\text{mass})(\Delta T)$$

Drop Units:

$$-[(4.184\ J/g^oC)(87\ g)(T_f - 72^oC)] = (4.184\ J/g^oC)(59\ g)(T_f - 13^oC)$$

$$-[(364.0)(T_f - 72^oC)] = (246.8)(T_f - 13^oC)$$

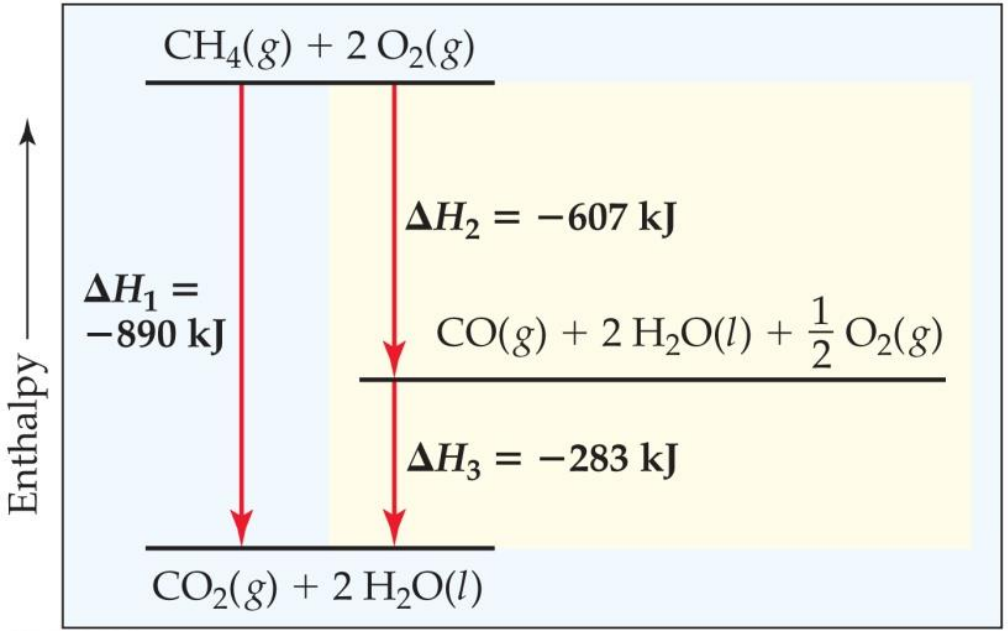
$$-364\ T_f + 26208 = 246.8\ T_f - 3208$$

$$29416 = 610.8\ T_f$$

$$T_f = 48.2^oC$$

Hess's Law

- ΔH is well known for many reactions, and it is inconvenient to measure ΔH for every reaction in which we are interested.
- However, we can estimate ΔH using published ΔH values and the properties of enthalpy.
- **Hess's law:** If a reaction is carried out in a series of steps, ΔH for the overall reaction will be equal to the sum of the enthalpy changes for the individual steps.



Hess's Law

Exp1: Find the standard molar enthalpy for the reaction

$$C_{(s)} + \frac{1}{2} O_{2(g)} \rightarrow CO_{(g)}$$

Given that $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H^o = -394$ kJ

and $CO_{2(g)} \rightarrow CO_{(g)} + \frac{1}{2} O_{2(g)}$ $\Delta H^o = +283$ kJ

Answer

1. Cancel

$C_{(s)} + O_{2(g)} \rightarrow \cancel{CO_{2(g)}}$ $\Delta H^o = -394$ kJ

$\cancel{CO_{2(g)}} \rightarrow CO_{(g)} + \frac{1}{2} O_{2(g)}$ $\Delta H^o = +283$ kJ

2. Add

$\mathbf{C_{(s)} + \frac{1}{2} O_{2(g)} \rightarrow CO_{(g)}}$ $\mathbf{\Delta H_o = -111}$ **kJ**

Exp2: The standard enthalpy changes for the formation of aluminium oxide and iron (III) oxide are

$$2\ Al_{(s)} + 3/2\ O_{2(g)} \rightarrow Al_2O_{3(s)} \qquad \Delta H^O = -1676\ kJ$$

$$2\ Fe_{(s)} + 3/2\ O_{2(g)} \rightarrow Fe_2O_{3(s)} \qquad \Delta H^O = -824\ kJ$$

Calculate ΔH^O for the reaction: $Fe_2O_{3(s)} + 2\ Al_{(s)} \rightarrow Al_2O_{3(s)} + 2\ Fe_{(s)}$

Answer

1. Inverse B

$$2\ Al_{(s)} + 3/2\ O_{2(g)} \rightarrow Al_2O_{3(s)} \qquad \Delta H^O = -1676\ kJ$$

$$Fe_2O_{3(s)} \rightarrow 2\ Fe_{(s)} + 3/2\ O_{2(g)} \qquad \Delta H^O = +824\ kJ$$

2. Cancel

$$2\ Al_{(s)} + 3/2\ \cancel{O_{2(g)}} \rightarrow Al2O_{3(s)} \qquad \Delta H^O = -1676\ kJ$$

$$Fe_2O_{3(s)} \rightarrow 2\ Fe_{(s)} + 3/2\ \cancel{O_{2(g)}} \qquad \Delta H^O = +824\ kJ$$

3. Add

$$\mathbf{Fe_2O_{3(s)} + 2\ Al_{(s)} \rightarrow Al_2O_{3(s)} + 2\ Fe_{(s)} \qquad \Delta H^o = -852\ kJ}$$

Exp3: Coal gasification converts coal into a combustible mixture of carbon monoxide and hydrogen, called coal gas, in a gasifier:

$$H_2O_{(l)} + C(s) \rightarrow CO_{(g)} + H_{2(g)} \qquad \Delta H^o = ?$$

Calculate the standard enthalpy change for this reaction from the following chemical equations:

$$2\ C_{(s)} + O_{2(g)} \rightarrow 2\ CO_{(g)} \qquad \Delta H^o = -222\ kJ$$

$$2\ H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)} \qquad \Delta H^o = -484\ kJ$$

$$H_2O_{(l)} \rightarrow H_2O_{(g)} \qquad \Delta H^o = +44\ kJ$$

Answer

1. Divide equation A and B by 2 and inverse equation B

$$C_{(s)} + \tfrac{1}{2}O_{2(g)} \rightarrow CO_{(g)} \qquad \Delta H^o = -111\ kJ$$

$$H_2O_{(g)} \rightarrow H_{2(g)} + \tfrac{1}{2}O_{2(g)} \qquad \Delta H^o = +242\ kJ$$

$$H_2O_{(l)} \rightarrow H_2O_{(g)} \qquad \Delta H^o = +44\ kJ$$

2. Cancel

$$C_{(s)} + \cancel{\tfrac{1}{2}O_{2(g)}} \rightarrow CO_{(g)} \qquad \Delta H^o = -111\ kJ$$

$$\cancel{H_2O_{(g)}} \rightarrow H_{2(g)} + \cancel{\tfrac{1}{2}O_{2(g)}} \qquad \Delta H^o = +242\ kJ$$

$$H_2O_{(l)} \rightarrow \cancel{H_2O_{(g)}} \qquad \Delta H^o = +44\ kJ$$

3. Add

$$\mathbf{H_2O_{(l)} + C_{(s)} \rightarrow CO_{(g)} + H_{2(g)}} \qquad \mathbf{\Delta H^o = +175\ kJ}$$